Regime switching forecasting for cryptocurrencies

Research | Open access | Published: 29 January 2025
Volume 7, pages 107–131, (2025) Cite this article

Abstract

There are many ways to model complex time series. The simplest approach is to increase the complexity, and thus, the flexibility of the model, for the entire time series. As an example, one could use a neural network. Another solution would be to change the parameters of a model dependent on the "state" or "regime" of the time series. A typical example here would be the Hidden Markov model (HMM). This paper combines the two concepts to create a Reinforcement Learning (RL) model that adds variables that depend on the state of the time series. To test the concept, the RL model is used with cryptocurrency data to determine the share to invest into the cryptocurrency index CRIX in order to maximize wealth. The results have shown that cryptocurrency metadata is useful as supplementary data for analysis of the respective prices. The Reinforcement learning model with regimes shows potential for investment management, but comes with some caveats.

Quantitative Finance
Stochastic Modelling
Cardiovascular models
Financial Time Series Prediction Using Machine Learning Techniques

1 Introduction

Asset returns behavior tends to change ever so often, making returns difficult to predict and portfolio management less consistent. This is especially true, if the asset class in question is based on cryptocurrencies. More complex models seem to have more success in the prediction process than the straightforward linear models, but they are difficult to interpret, and thus may not be replicable in a different dataset or environment.

The second step trains three Reinforcement Learning models to determine the portfolio weights. One of them will have regime probabilities (the raw output of the previous step), the second one will only include the predicted regime (the regime with the highest probability), as well as a reference one with no regimes. This step uses the hourly data for the CRIX cryptocurrency index, also provided by the BRC.

The remainder of the paper will be structured as follows. Section 2 gives a literature overview. Then, Sect. 3 describes the methodology for both steps of the model. Section 4 shows the model configuration in practice for BTC analysis and CRIX trading. For the first step, it is evaluated, how the weighting changes the regime prediction, and how well both regimes in general as well as regime changes are predicted. Then, LIME is used to determine, which of the variables had the strongest influence on the decision making. For the second step, the trading of CRIX with the three different methods is evaluated and compared. Afterwards, the results are discussed in more detail and a conclusion is drawn. All

Another architecture that is often used with time series is the Convolutional Neural Network (CNN) (Lecun et al., 1998). While primarily used for image data, CNNs have been adapted for time series by treating the data as a one-dimensional "image". This allows them to capture local patterns and temporal dependencies within the data. Other techniques include Time Delay Neural Networks (Waibel et al., 1989) and Attention Mechanisms (Vaswani et al., 2017).

Reinforcement learning is a framework designed to learn the optimal actions over a period with the goal to maximize the reward over that period. It is often used in tandem with neural networks, as fully modeling Q-tables is often not an option. Q-learning is the simplest method, and it has many variations, such as Deep Q-Learning (DQN), Double DQN (Hasselt et al., 2016) and Dueling DQN (Wang et al., 2016). Common actor-critic algorithms are Deep Deterministic Policy Gradient (DDPG) (Lillicrap et al., 2016), Advantage Actor Critic (A2C) (Mnih et al., 2016) and Proximal Policy Optimization (PPO) (Schulman et al., 2017).

3 Methodology

3.1 Data overview and preprocessing

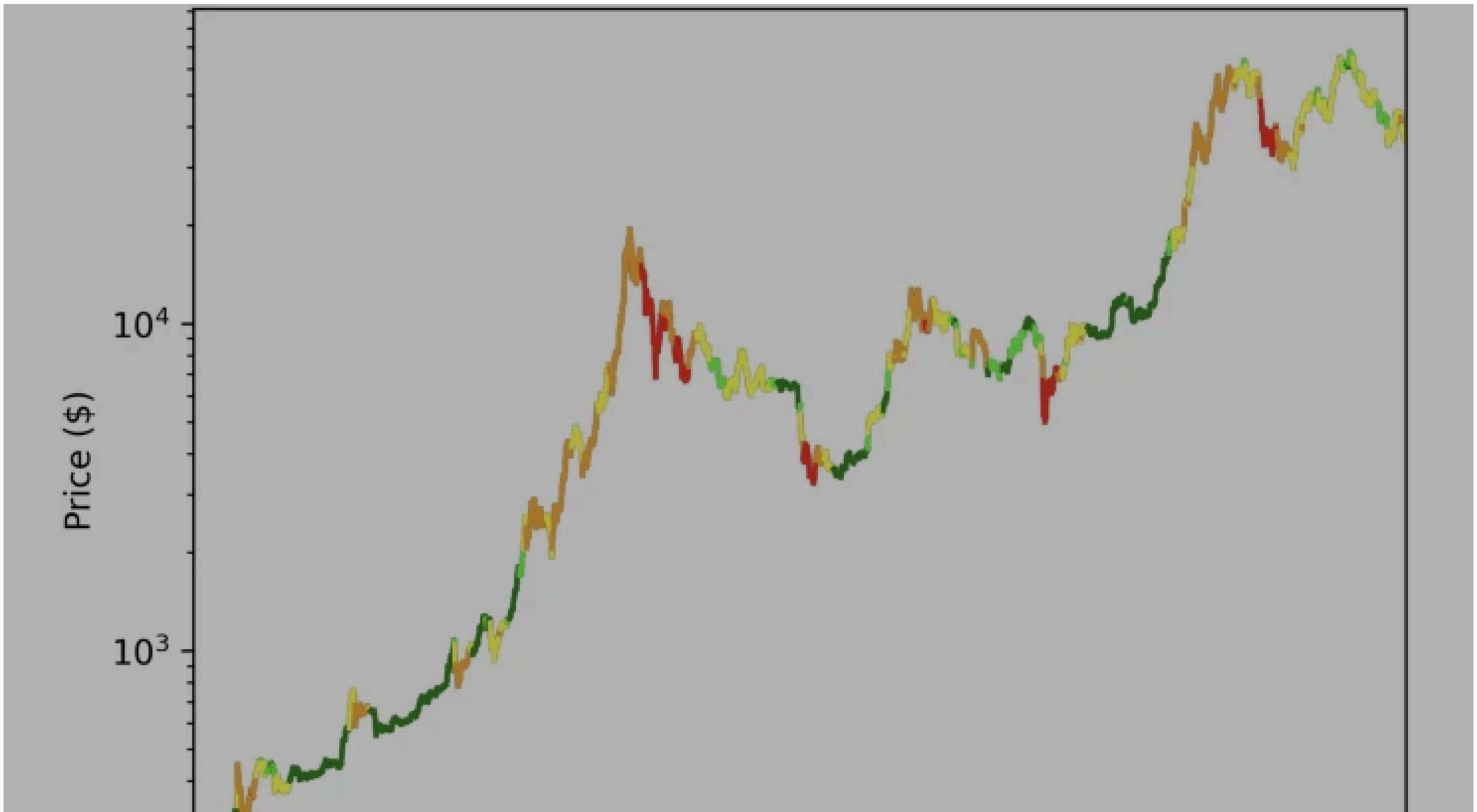
The CRIX was initially developed by Härdle and Trimborn (2015) and further studied by Trimborn and Härdle (2018), Kim et al. (2021) and Hou et al. (2019). At the time of writing, the CRIX is provided by Royalton Partners.

The dominant cryptocurrency of the CRIX is Bitcoin (BTC), and the other coins in the index are highly positively correlated with BTC (Candila, 2021), so, to simplify the calculations, only BTC metadata is used. The dataset ranges from 07.09.2015 to 31.05.2022 and is recorded daily. The data are extracted from the Blockchain Research Center[3] (BRC) and is listed in Table 1.

The distress regime includes return quantiles p_r as a additional criterion. It was necessary to introduce it, as, unlike most assets, cryptocurrencies do not have a significantly negative skewness. In fact, the skewness of BTC in during the combined training and test sets is almost zero ($-$0.0362). Thus, very high volatility in cryptocurrencies have a similar probability to result in positive returns as negative returns (Table 2).

Table 2 Regime criteria

Overall, distress is determined using a distress score p_d, which is a combination of the volatility (p_σ) and return quantiles (p_r):



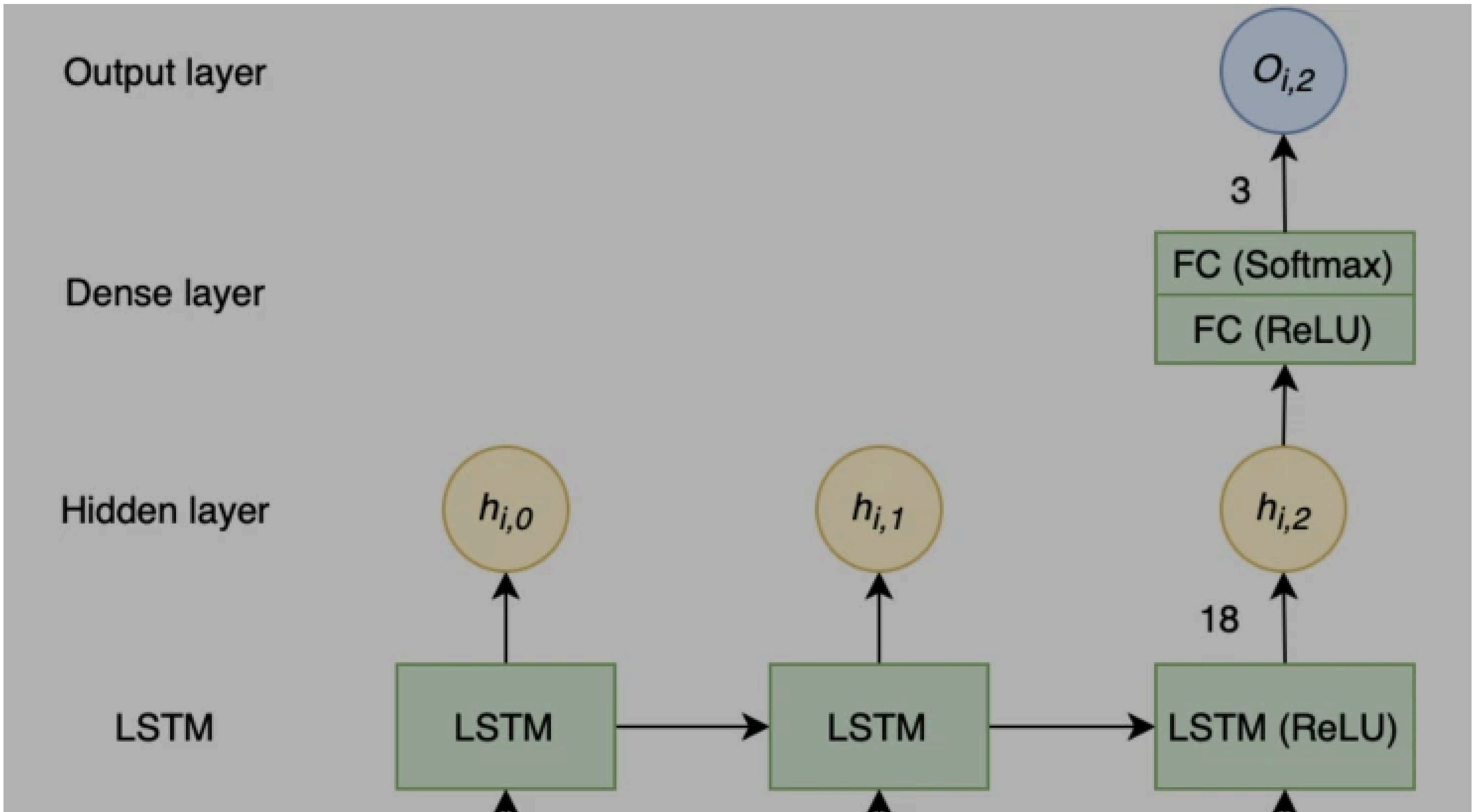
10^4
10^3
Price ($)

Denote $g \in G, G \overset{\operatorname{def}}{=}\{1, 2, 3\}$ as a regime in the regime space, and g_t is the regime at time *t*. Then, in order to account for the imbalance, the weight w_t is introduced when calculating the loss function:

$$\begin{aligned} w_t=(1-\lambda)\frac{T}{\sum _{s=1}^{T} I\{g_t=g_s\}}+\lambda \frac{\sum _{s=1}^{T} I\{g_s\ne g_{s-1}\}}{\sum _{s=1}^{T} I\{g_t=g_s\}I\{g_s\ne g_{s-1}\}} \end{aligned}$$

(2)

where T is the total length of the time series in the training set and is a hyperparameter. Here, the first term corrects for the imbalance in the data and the second adds more weight to data points, where the regime switches. $\lambda \in [0,1]$ determines, how important regime-switching data points relative to other data points are.



Output layer
$O_{i,2}$
3
Dense layer
FC (Softmax)
FC (ReLU)
Hidden layer
$h_{i,0}$
$h_{i,1}$
$h_{i,2}$
18
LSTM
LSTM
LSTM
LSTM (ReLU)

$$\begin{aligned} l=-\sum _{t=1}^{T}w_t y_t \log \hat{y}_t \end{aligned}$$

(3)

In order to evaluate variable importance, the LIME algorithm is applied (Ribeiro et al. 2016). The objective of the algorithm is to ensure both interpretability and local fidelity. This is achieved by solving the minimization problem:

$$\begin{aligned} \xi (x)=\arg \min _{g\in G}L(f,g,\pi _x)+\Omega (g) \end{aligned}$$

(4)

where *L* measures how close the explanation is to the prediction of the original model *f*, $\Omega (g)$ is the penalty for model complexity, and $\pi _x$ is the magnitude of neighbourhood around instance *x*

$$\begin{aligned} & L(s,a,\theta _k,\theta)=\min \Bigg (\frac{\pi _{\theta }(a|s)}{\pi _{\theta _k}(a|s)} A_{\pi _{\theta _k}}(s,a), g(\epsilon , A_{\pi _{\theta _k}})\Bigg) \end{aligned}$$

(5)

$$\begin{aligned} & g(\epsilon ,A)= {\left\{ \begin{array}{ll} (1+\epsilon)A, A\ge 0\\ (1-\epsilon)A, A<0 \end{array}\right. } \end{aligned}$$

(6)

Here, a—action, s—state, $\pi _{\theta }$—current policy, $\pi _{\theta _k}$—previous policy, A—advantage, $A_{\pi _{\theta _k}}$—advantage of the previous policy, ϵ—hyperparameter used to regulate clipping range.

- Relative strength index (RSI) (Wilder, 1978)

$$\begin{aligned} RSI = 100-\left[100/\left(1+\frac{n_{up}}{n_{down}}\right) \right] \end{aligned}$$

(8)

where n_{up}—average gain during upside periods and n_{down}—average gain during downside periods

- Commodity channel index (CCI) (Schlossberg, 2006)

$$\begin{aligned} CCI = \frac{1}{0.015}\frac{p_t - MA(p_t)}{MD(p_t)} \end{aligned}$$

(9)

There are two ways to get w_{t} as a result:

- Winner-takes-all—the regime g_t with the highest probability will have a value of 1. For example, if regime "Low" is predicted, the input vector is (0, 0, 1).
- Weighted average—the values are probabilities of a regime being in place. This is essentially the raw output of the regime prediction stage.

Both of these options will be examined as separate models. In addition, a third version without any regimes is tested as a reference.

After the actions are received, the reward is calculated:

$$\begin{aligned} C_{t}= & a_t \times B_{t-1} / P_{t-1} \end{aligned}$$

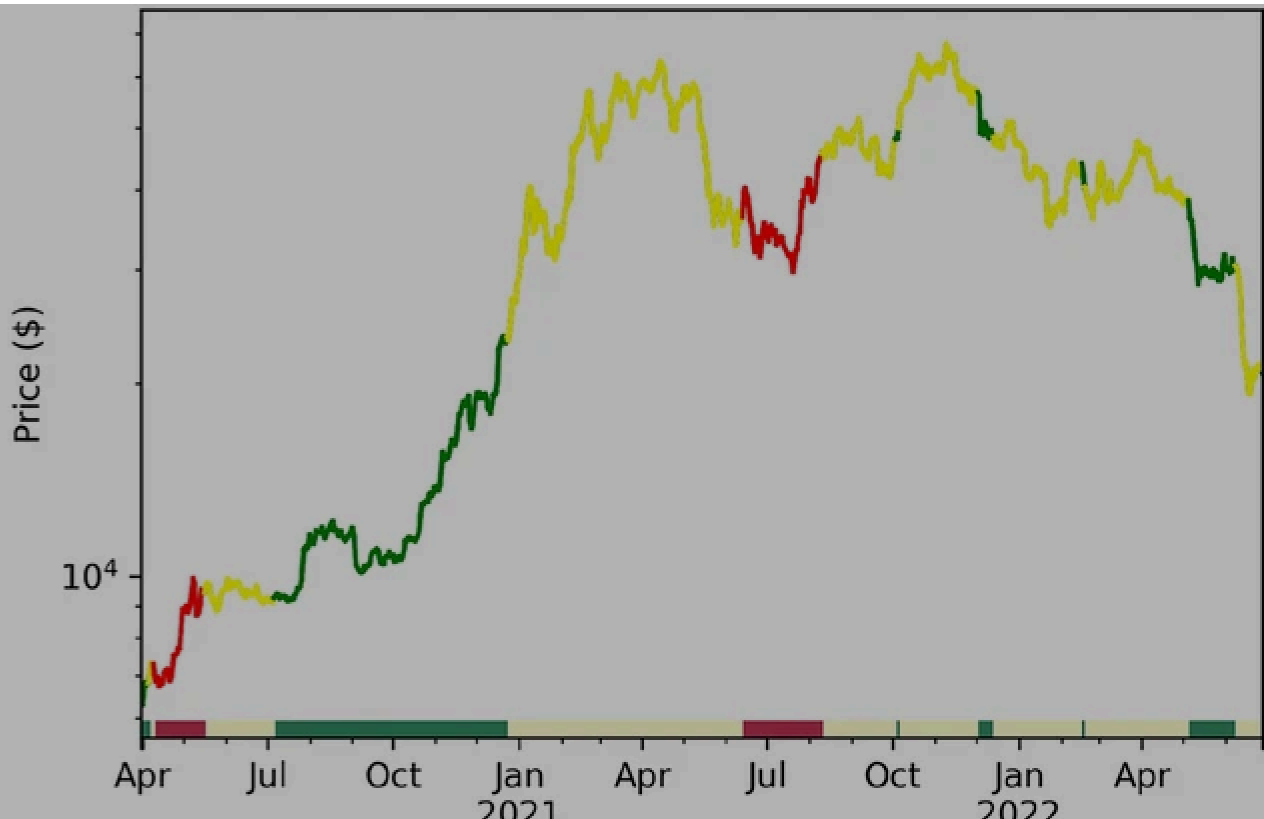
4.1.1 Performance

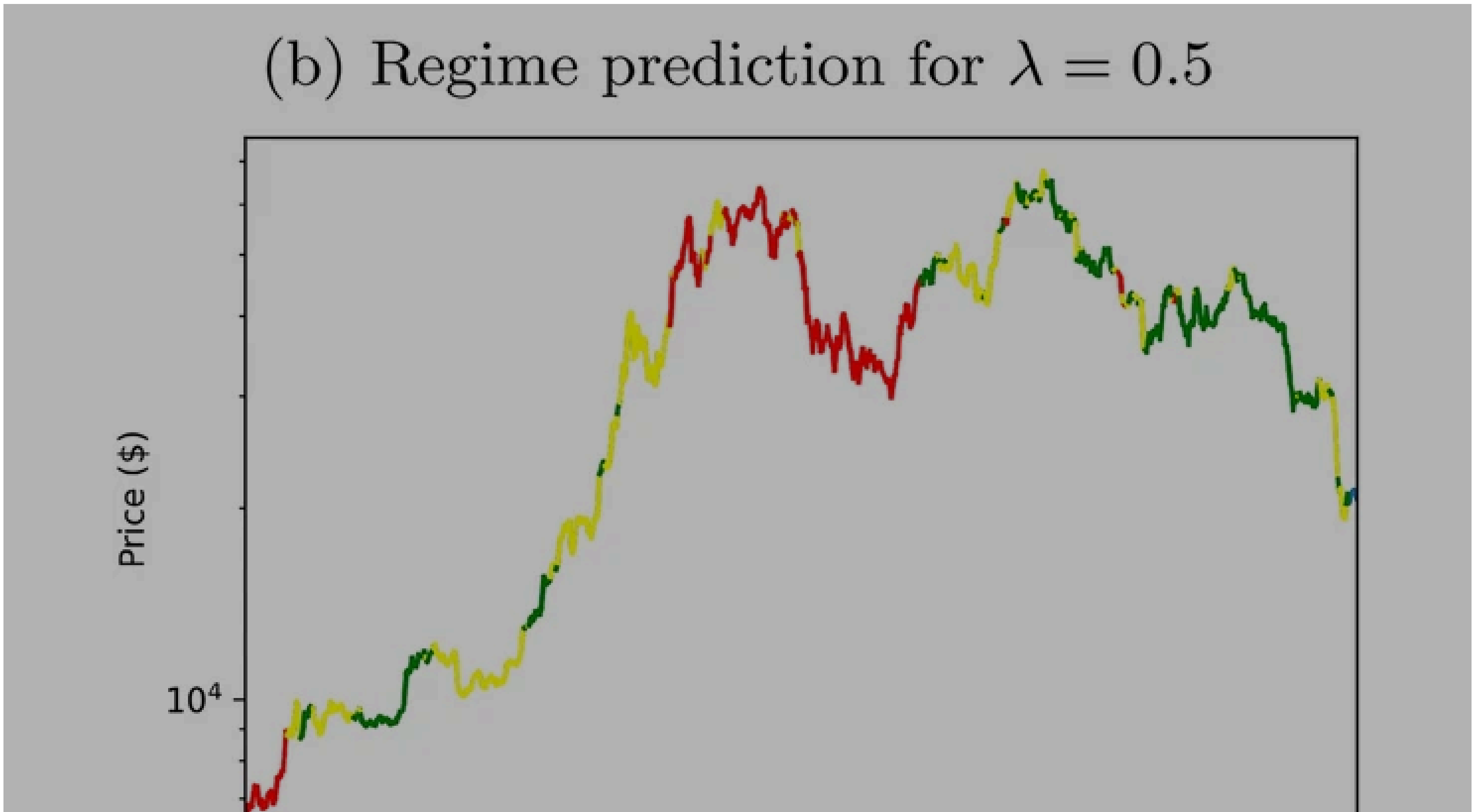
At this stage, the data are split a follows:

- Training set: 08.08.2015–10.12.2019
- Test set: 11.12.2019–27.02.2022

The results for BTC are presented in the following tables for various metrics. As a reference for the main LSTM with weights, results for LSTM without weights and the ARMA-GARCH model are presented (Table 4):

Table 4 F1-Score for different methods and \(\lambda\)


Price ($)
10^4
Apr
Jul
Oct
Jan 2021
Apr
Jul
Oct
Jan 2022
Apr


(b) Regime prediction for $\lambda = 0.5$
Price ($)
10^4

(d) Regime prediction for $\lambda = 1$

Bitcoin price plot with regimes. Regime breakdown: zone 2, distress. RRL_Regime-Prediction

4.1.2 Interpretation

The model will be interpreted using LIME (Local Interpretable Model-Agnostic Explanations) (Ribeiro et al., 2016). LIME is a local surrogate model, i.e. it approximates the predictions of the black box model with an interpretable one. To determine the impact for each variable \(\xi (x)\), values are perturbed, forming a new dataset, then the interpretable model is fit, such that the loss function L is minimized:

(17)

To determine the most influential variables, the LIME-values were calculated for every data point in the test set. A sample for a single data point can be found in Appendix B.

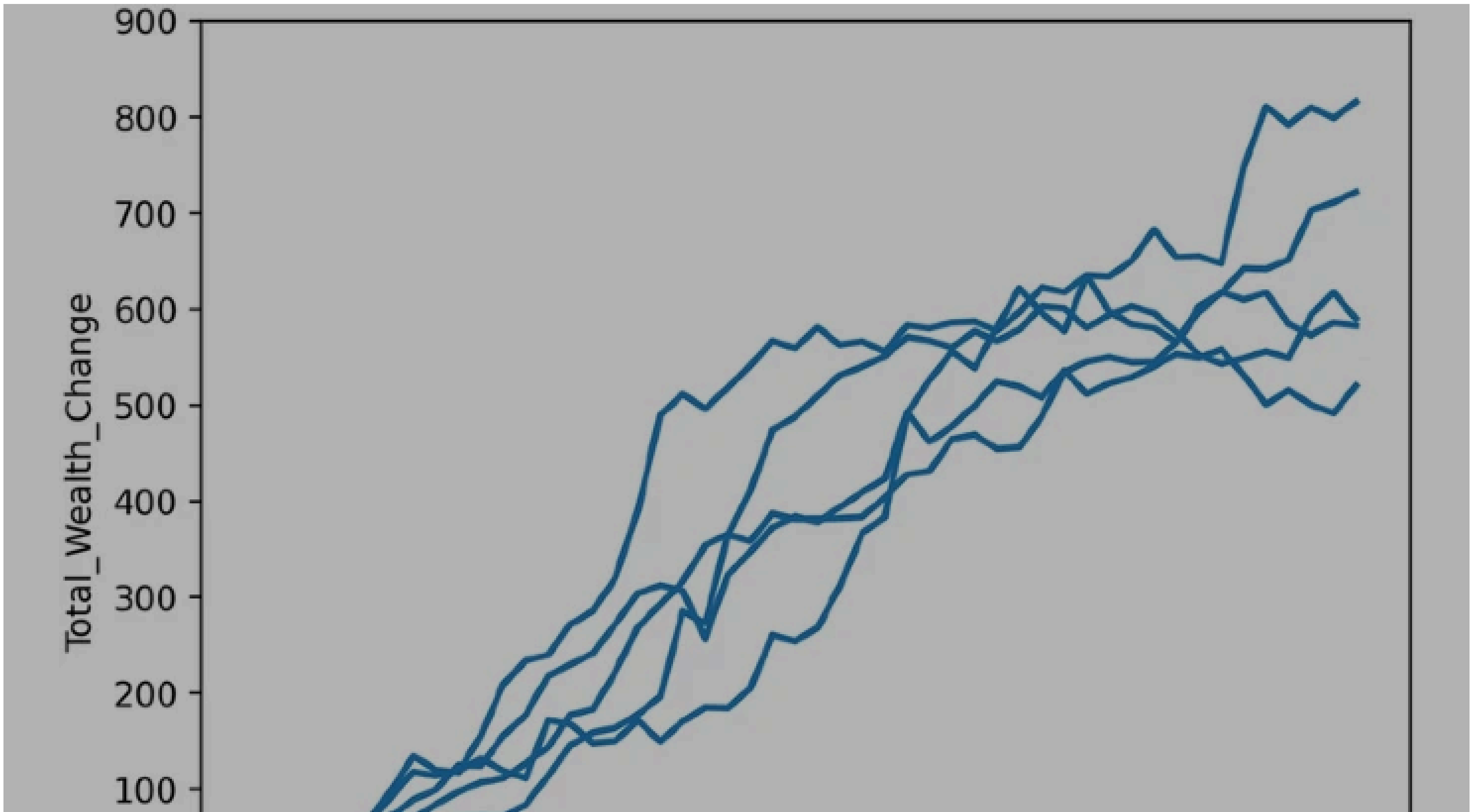
For an approximation of the overall influence, the variables are ranked based on the absolute values (with rank 1 being the most influential one) for every data point in the test set. The mean of the values is then calculated for the final ranking, which can be observed in Appendix B.

It it fairly obvious that previous regimes, price, volatility and drawdown all have a considerable influence over the current regime. However, metadata variables, such as sent coins, average transaction fee per day, average transaction value per day, number of unique addresses per day and average fee to reward. These variables mostly relate to trading volume

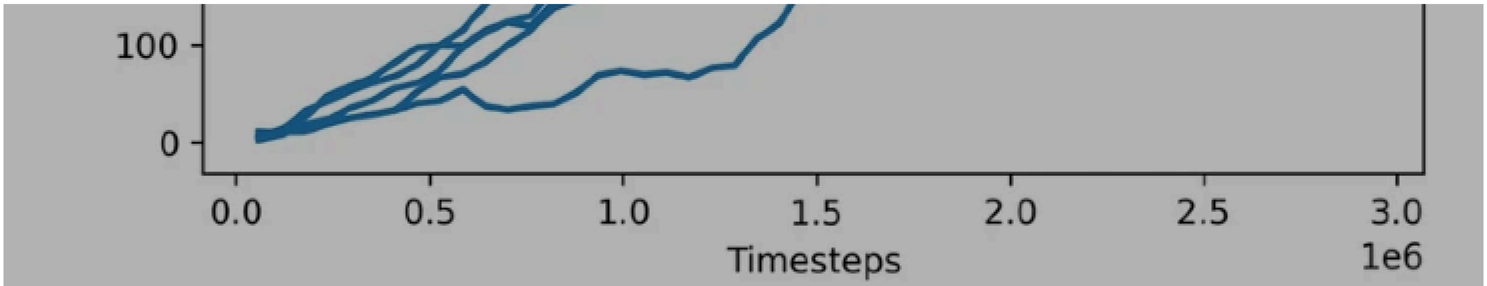
- Training set: 06.03.2020–08.07.2021
- Validation set: 09.07.2021–19.12.2021
- Test set: 20.12.2021–31.05.2022

Then, for all three variants of the model, the rewards are tracked as they are trained. They can be observed in Fig. 4. As the training process is stochastic, each variant is run five times to make sure the performance is consistent.

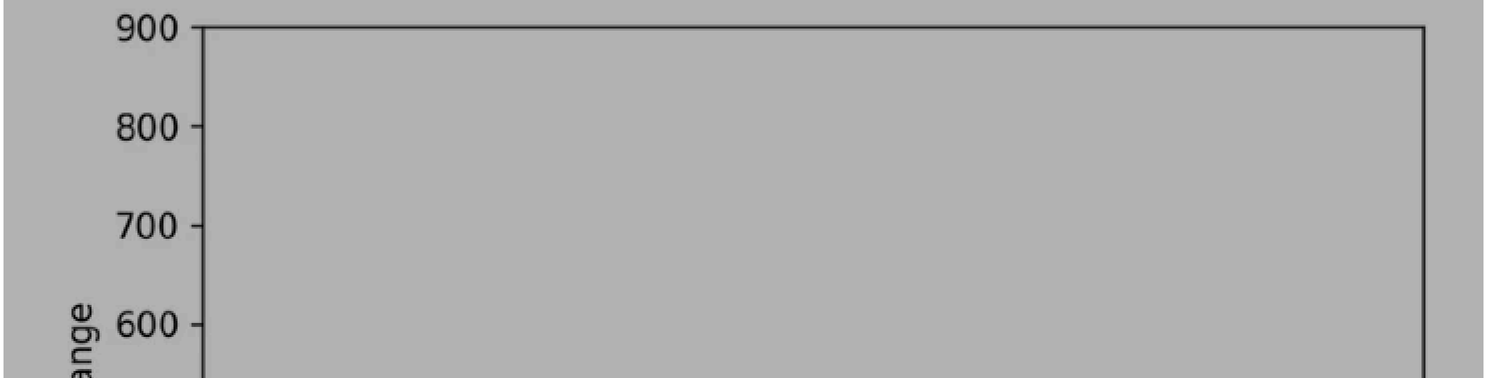
Fig. 4



Total_Wealth_Change
900
800
700
600
500
400
300
200
100



(b) Regime predictions



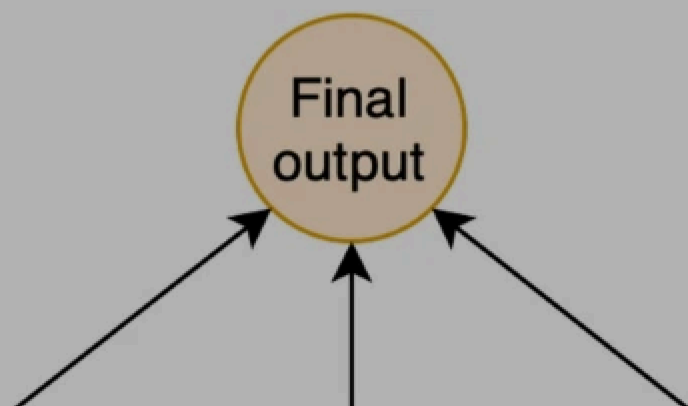
At least in the training set, the additional regime data does make a difference. Both the predicted regime probabilities and regime values significantly increase the reward compared to the “no-regimes” variant for the same training period, so it can be stated that the regimes do contain meaningful information for investment purposes.

As for the performance in the validation and test sets, the difference there is rather insignificant. The performance metrics for the training, validation and test sets can be observed in Table 6, while the reward development over time is shown in Appendix C.

Unfortunately, it seems like the non-stationary nature complicates any kind of prediction task. Of course, it could just be, that regime predictions happened to be beneficial for the task in one time period, but not the other. It could therefore be a case of insufficient data, as regime changes are uncommon during the given time period.

architectures could be used. One example would be to split the network in a way that each network would train for a separate regime (see Fig. 5). To make the current version of the model more comparable to a buy-and-hold strategy and expand on its ideas, trading costs, leverage and short-selling could be introduced.

Fig. 5



Sample architecture of the policy learning network

6 Conclusion

A new approach to analyze the cryptocurrency market was introduced and applied to Bitcoin (BTC). Regimes were introduced that could be useful for further analysis. Using a weight system to counteract imbalance, a method was shown that could effectively predict future regimes with a high emphasis on when the regime switches. The LIME method was used to gain insight to what variables from the metadata had the most impact on the outcome. Those variables ended up being related to trading volume.

Afterwards, the regime prediction was used as part of a dataset in a reinforcement learning

1. https://quantlet.com.

2. https://quantinar.com.

3. https://blockchain-research-center.com.

References

Appel, G. (2005). *Technical analysis: Power tools for active investors*. FT Press.

Google Scholar

Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). Learning phrase representations using RNN encoder–decoder for statistical machine translation. In A. Moschitti, B. Pang, & W. Daelemans (Eds.), *Proceedings of the 2014 conference on empirical methods in natural language processing (EMNLP)* (pp. 1724–1734). Association for Computational Linguistics.

Chapter Google Scholar

Härdle, W.K., & Trimborn, S. (2015). Crix or evaluating blockchain based currencies. Technical report. SFB 649 discussion paper.

Hasselt, Hv., Guez, A., & Silver, D. (2016). Deep reinforcement learning with double q-learning. *Proceedings of the thirtieth AAAI conference on artificial intelligence* (pp. 2094–

Kim, A., Trimborn, S., & Härdle, W. K. (2021). Vcrix—a volatility index for crypto-currencies. *International Review of Financial Analysis, 78*, 101915.

Article Google Scholar

Lecun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE, 86*, 2278–2324. https://doi.org/10.1109/5.726791

Article Google Scholar

Li, Y., Zheng, W., & Zheng, Z. (2019). Deep robust reinforcement learning for practical algorithmic trading. *IEEE Access*, 7, 108014–108022. https://doi.org/10.1109/ACCESS.2019.2932789

Article Google Scholar

Maringer, D., & Ramtohul, T. (2012). Regime-switching recurrent reinforcement learning for investment decision making. *Computational Management Science*, *9*, 89–107. https://doi.org/10.1007/s10287-011-0131-1

Article Google Scholar

Mnih, V., Badia, A. P., Mirza, M., Graves, A., Lillicrap, T., Harley, T., Silver, D., & Kavukcuoglu, K. (2016). Asynchronous methods for deep reinforcement learning. In M. F. Balcan & K. Q. Weinberger (Eds.), *Proceedings of the 33rd international conference on machine learning* (pp. 1928–1937). New York, New York, USA: PMLR.

Google Scholar

Google Scholar

Sattarov, O., Muminov, A., Lee, C. W., Kang, H. K., Oh, R., Ahn, J., Oh, H. J., & Jeon, H. S. (2020). Recommending cryptocurrency trading points with deep reinforcement learning approach. *Applied Sciences*. https://doi.org/10.3390/app10041506

Article Google Scholar

Schlossberg, B. (2006). *Technical analysis of the currency market*. Wiley Trading.

Google Scholar

Schulman, J., Wolski, F., Dhariwal, P., Radford, A., & Klimov, O. (2017). Proximal policy

Article Google Scholar

Wang, Z., Schaul, T., Hessel, M., Van Hasselt, H., Lanctot, M., & De Freitas, N. (2016). Dueling network architectures for deep reinforcement learning. In *Proceedings of the 33rd international conference on international conference on machine learning* (Vol. 48, pp. 1995–2003).

Wilder, J. (1978). New concepts in technical trading systems. Trend Research. https://books.google.de/books?id=WesJAQAAMAAJ.

Funding

Ilyas Agakishiev has done: data analysis, coding, Wolfgang Karl Härdle proposed the regime switching framework, Denis Becker implemented the data analysis and Xiaorui ZUO has put the research elements together and assisted in coding the reinforcement training

Corresponding author

Correspondence to Wolfgang Karl Härdle.

Ethics declarations

Conflict of interest

The authors declare no conflict of interest.

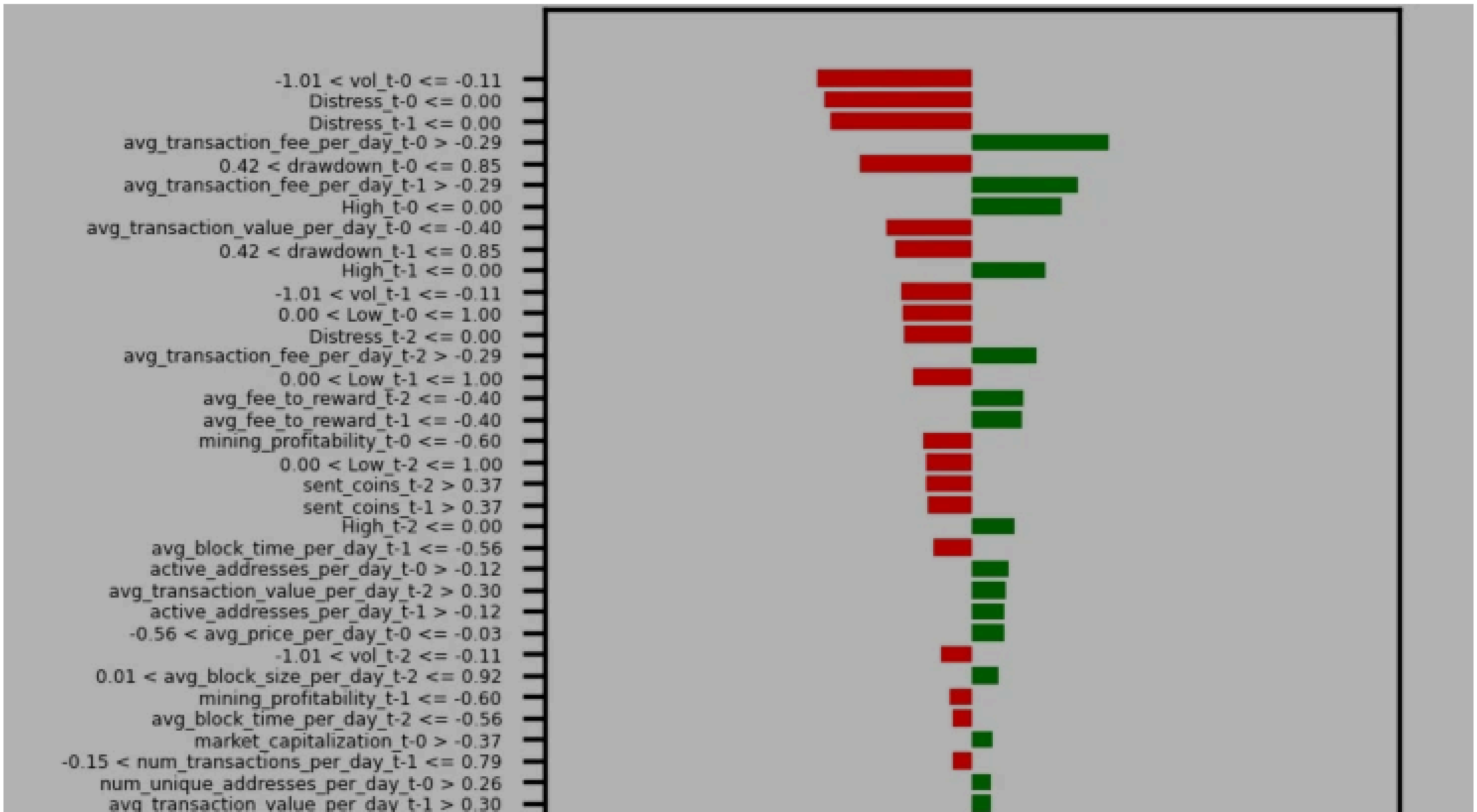
See Table 7.

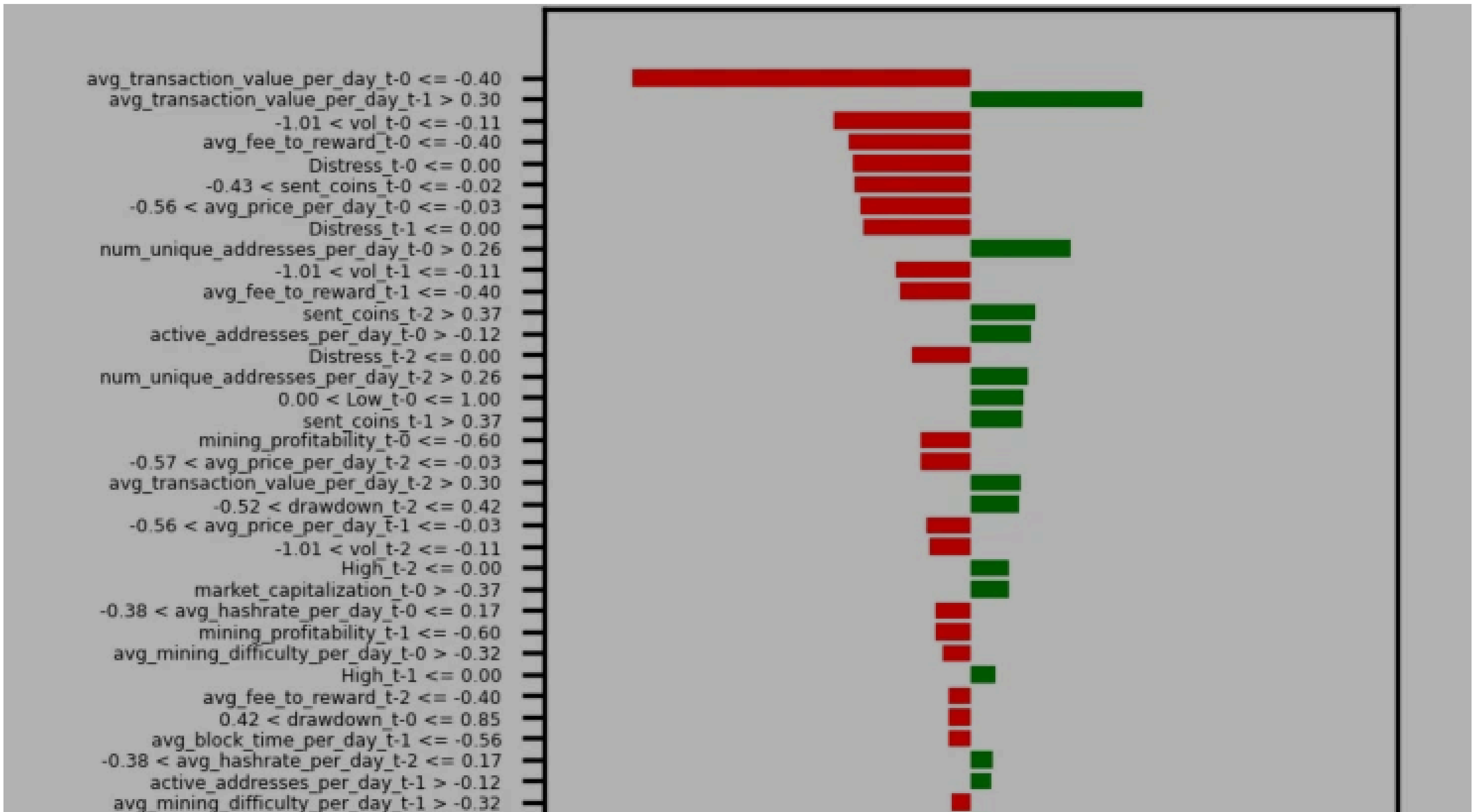
Table 7 Descriptive statistics of the Bitcoin metadata

1.2 B. LIME analysis

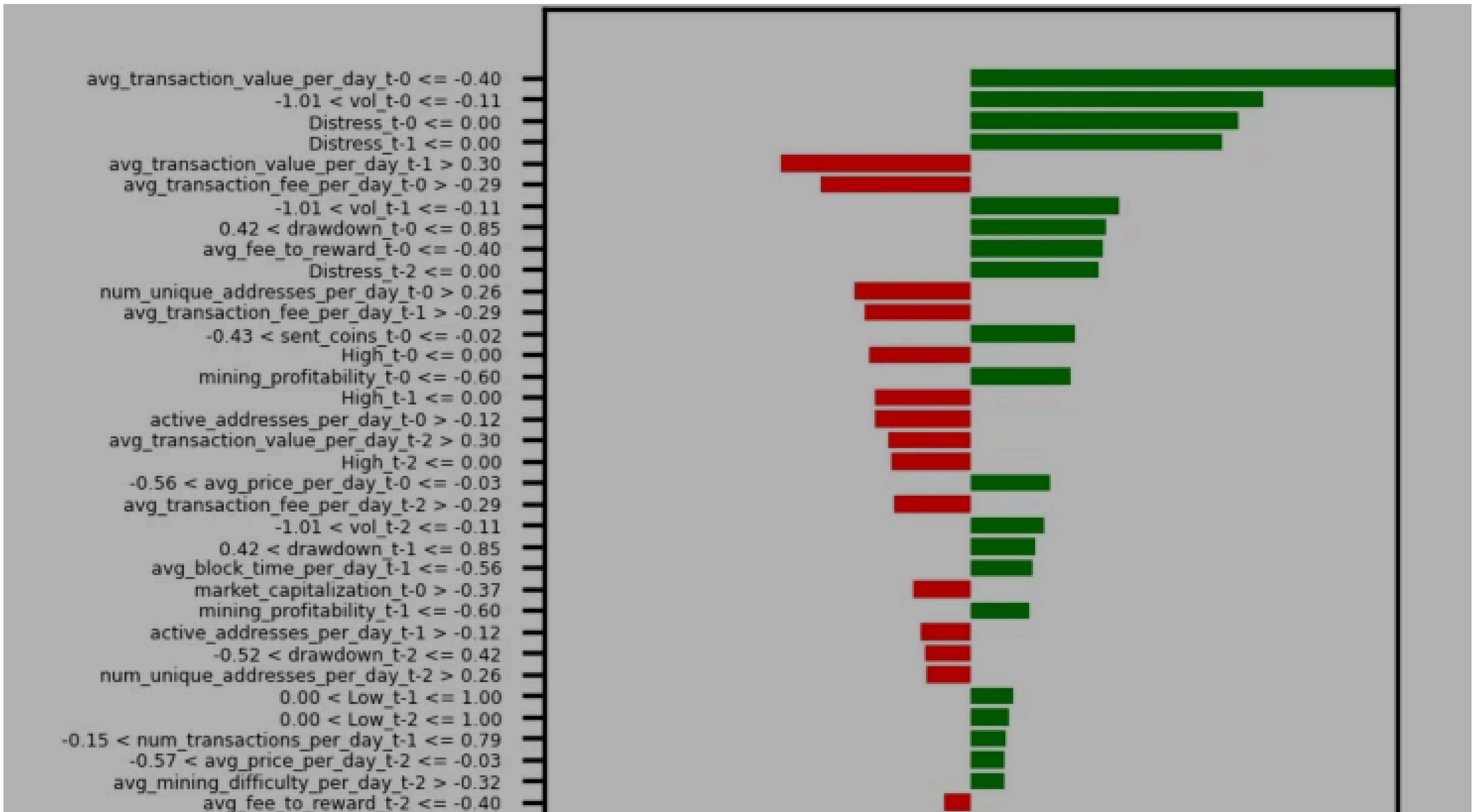
See Figs. 6, 7, 8 and Table 8.

Fig. 6


-1.01 < vol_t-0 <= -0.11
Distress_t-0 <= 0.00
Distress_t-1 <= 0.00
avg_transaction_fee_per_day_t-0 > -0.29
0.42 < drawdown_t-0 <= 0.85
avg_transaction_fee_per_day_t-1 > -0.29
High_t-0 <= 0.00
avg_transaction_value_per_day_t-0 <= -0.40
0.42 < drawdown_t-1 <= 0.85
High_t-1 <= 0.00
-1.01 < vol_t-1 <= -0.11
0.00 < Low_t-0 <= 1.00
Distress_t-2 <= 0.00
avg_transaction_fee_per_day_t-2 > -0.29
0.00 < Low_t-1 <= 1.00
avg_fee_to_reward_t-2 <= -0.40
avg_fee_to_reward_t-1 <= -0.40
mining_profitability_t-0 <= -0.60
0.00 < Low_t-2 <= 1.00
sent_coins_t-2 > 0.37
sent_coins_t-1 > 0.37
High_t-2 <= 0.00
avg_block_time_per_day_t-1 <= -0.56
active_addresses_per_day_t-0 > -0.12
avg_transaction_value_per_day_t-2 > 0.30
active_addresses_per_day_t-1 > -0.12
-0.56 < avg_price_per_day_t-0 <= -0.03
-1.01 < vol_t-2 <= -0.11
0.01 < avg_block_size_per_day_t-2 <= 0.92
mining_profitability_t-1 <= -0.60
avg_block_time_per_day_t-2 <= -0.56
market_capitalization_t-0 > -0.37
-0.15 < num_transactions_per_day_t-1 <= 0.79
num_unique_addresses_per_day_t-0 > 0.26
avg_transaction_value_per_day_t-1 > 0.30



avg_transaction_value_per_day_t-0 <= -0.40
avg_transaction_value_per_day_t-1 > 0.30
-1.01 < vol_t-0 <= -0.11
avg_fee_to_reward_t-0 <= -0.40
Distress_t-0 <= 0.00
-0.43 < sent_coins_t-0 <= -0.02
-0.56 < avg_price_per_day_t-0 <= -0.03
Distress_t-1 <= 0.00
num_unique_addresses_per_day_t-0 > 0.26
-1.01 < vol_t-1 <= -0.11
avg_fee_to_reward_t-1 <= -0.40
sent_coins_t-2 > 0.37
active_addresses_per_day_t-0 > -0.12
Distress_t-2 <= 0.00
num_unique_addresses_per_day_t-2 > 0.26
0.00 < Low_t-0 <= 1.00
sent_coins_t-1 > 0.37
mining_profitability_t-0 <= -0.60
-0.57 < avg_price_per_day_t-2 <= -0.03
avg_transaction_value_per_day_t-2 > 0.30
-0.52 < drawdown_t-2 <= 0.42
-0.56 < avg_price_per_day_t-1 <= -0.03
-1.01 < vol_t-2 <= -0.11
High_t-2 <= 0.00
market_capitalization_t-0 > -0.37
-0.38 < avg_hashrate_per_day_t-0 <= 0.17
mining_profitability_t-1 <= -0.60
avg_mining_difficulty_per_day_t-0 > -0.32
High_t-1 <= 0.00
avg_fee_to_reward_t-2 <= -0.40
0.42 < drawdown_t-0 <= 0.85
avg_block_time_per_day_t-1 <= -0.56
-0.38 < avg_hashrate_per_day_t-2 <= 0.17
active_addresses_per_day_t-1 > -0.12
avg_mining_difficulty_per_day_t-1 > -0.32

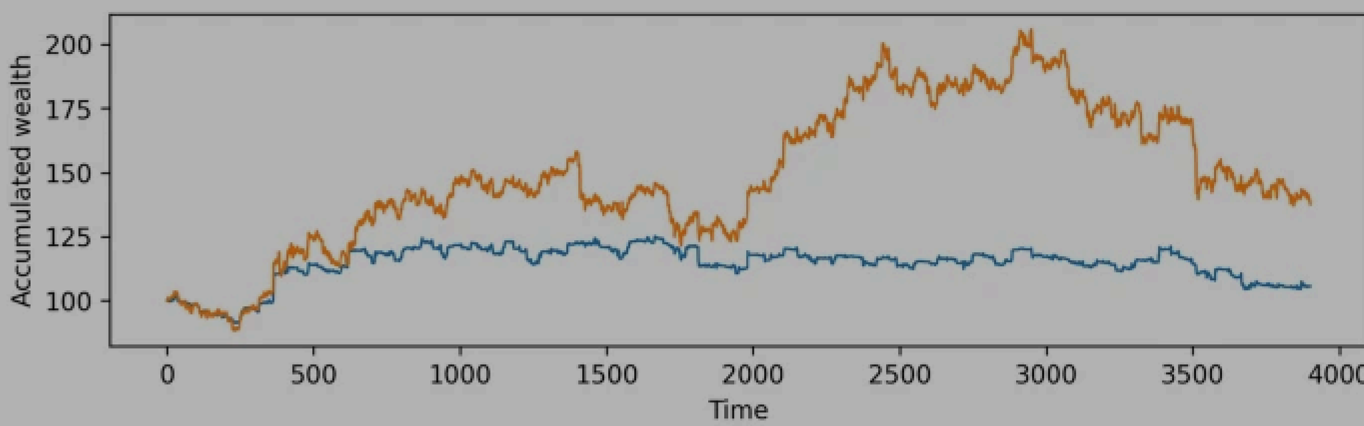


avg_transaction_value_per_day_t-0 <= -0.40
-1.01 < vol_t-0 <= -0.11
Distress_t-0 <= 0.00
Distress_t-1 <= 0.00
avg_transaction_value_per_day_t-1 > 0.30
avg_transaction_fee_per_day_t-0 > -0.29
-1.01 < vol_t-1 <= -0.11
0.42 < drawdown_t-0 <= 0.85
avg_fee_to_reward_t-0 <= -0.40
Distress_t-2 <= 0.00
num_unique_addresses_per_day_t-0 > 0.26
avg_transaction_fee_per_day_t-1 > -0.29
-0.43 < sent_coins_t-0 <= -0.02
High_t-0 <= 0.00
mining_profitability_t-0 <= -0.60
High_t-1 <= 0.00
active_addresses_per_day_t-0 > -0.12
avg_transaction_value_per_day_t-2 > 0.30
High_t-2 <= 0.00
-0.56 < avg_price_per_day_t-0 <= -0.03
avg_transaction_fee_per_day_t-2 > -0.29
-1.01 < vol_t-2 <= -0.11
0.42 < drawdown_t-1 <= 0.85
avg_block_time_per_day_t-1 <= -0.56
market_capitalization_t-0 > -0.37
mining_profitability_t-1 <= -0.60
active_addresses_per_day_t-1 > -0.12
-0.52 < drawdown_t-2 <= 0.42
num_unique_addresses_per_day_t-2 > 0.26
0.00 < Low_t-1 <= 1.00
0.00 < Low_t-2 <= 1.00
-0.15 < num_transactions_per_day_t-1 <= 0.79
-0.57 < avg_price_per_day_t-2 <= -0.03
avg_mining_difficulty_per_day_t-2 > -0.32
avg_fee_to_reward_t-2 <= -0.40

1.3 C. Trading results

See Figs. 9 and 10.

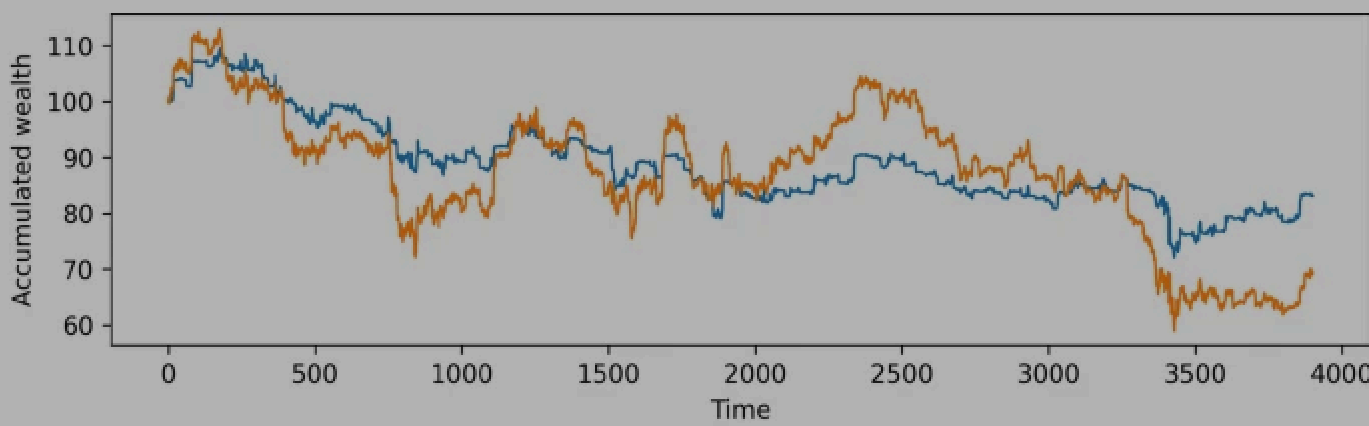
Fig. 9



(a) Regime prediction probabilities

Validation set

Fig. 10



Test set. RRL_Reinforcement-Learning

Rights and permissions

DOI

https://doi.org/10.1007/s42521-024-00123-2

Share this article

Anyone you share the following link with will be able to read this content:

Get shareable link

Provided by the Springer Nature SharedIt content-sharing initiative

Keywords

Regime switching | Machine learning | Crypto currencies | Reinforcement learning | FinTech